Laxman Acharya

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Summary

Computer Engineering student at Kathmandu University building full-stack, mobile, AI-assisted, and systems software across healthcare, document retrieval, GIS civic response, and Linux kernel fixes. Hands-on experience with TypeScript, Python, C/C++, React/Vite, FastAPI, Node.js, SQL, Docker, and Linux. Interested in scalable software engineering, system design, and open-source infrastructure.

Selected Work

PaperSorted 2026
Static Search / Document Retrieval

- Built a static Kathmandu University exam-paper archive with subject search, program-aware browsing, and inline PDF viewing across **561 subjects** and **2,792 papers**.
- Modeled **17 program tracks** and **138 semester groupings**; generated precomputed search indexes for fast backend-free retrieval and structured discovery.

HamroCare
Healthcare Systems / Mobile Application

- Developed a mobile healthcare application with AI-assisted symptom checking, family health tracking, and longitudinal medical-interaction history.
- Designed responsive interfaces and integrated REST APIs for real-time creation, retrieval, and management of healthcare data.

Zero-Burn OS 2026
GIS / Civic Response / Environmental Technology

- Built a GIS and phone-IoT civic response system that turns citizen QR reports and geotagged evidence into mapped, prioritized open-waste-burning hotspots.
- Implemented explainable risk scoring, duplicate clustering, route prioritization, before/after resolution evidence, offline demo data, and printable action reports.

BridgeLens
Language Technology / Browser Extension

- Developed a browser extension supporting translation among English, Nepali, and Tamang to improve multilingual access for local users.
- Implemented extension workflows with modern browser APIs and a lightweight JavaScript interface for in-context translation.

Mulyankan
Academic Software / Object-Oriented Design

- Contributed educational-software features using object-oriented design and database integration; collaborated on implementation, debugging, and functional testing.

Upstream Contributions

- **Linux kernel** | Submitted accepted patches across Bluetooth, IIO, and DRM/panel drivers; fixes covered firmware-bound validation and runtime-PM/regulator cleanup paths, with maintainer review and stable marking in 2026.

Recognition & Achievements

- **HackForUHC 2026** | Won 1st place as team CLAIM∅ at HackForUHC 2026, held June 18-19, 2026; invited to present our solution at the 10th anniversary openIMIS Community Meeting, Kathmandu (June 23-25, 2026).
- **Harvard Health Systems Innovation Lab Hackathon** | Participated with team Never Silent Pass in the 7th edition at the Kathmandu Hub, focused on building high-value health systems through AI (April 10-11, 2026).

Education

Kathmandu University B.E. in Computer Engineering | Expected 2028

Skills

Languages: TypeScript, Python, C/C++, JavaScript, SQL, Verilog
Systems: React/Vite, Node.js, Express, PostgreSQL, Redis, MongoDB, Docker, Linux Kernel
Tooling: Socket.IO, libsodium, KUnit, EasyOCR, Tesseract, Icarus Verilog, GTKWave, Git, Make